

Enhanced Machine Learning Approach for Accurate Heart Disease Diagnosis

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Abstract

Heart disease is a problem that causes a lot of deaths around the world. We need to find ways to predict it accurately and early. This study uses a machine learning approach with the K-Nearest Neighbour algorithm to predict heart disease. We used information like age, blood pressure and cholesterol levels to make predictions. We had to clean up the data by filling in missing values and making sure all the numbers were on the same scale. This helped our model work better. We tried settings for the K-Nearest Neighbour algorithm to find the best one. We tested it times to make sure it was reliable.

We tried numbers for K and different ways of measuring distance to find what worked best. The results were good. Our model was about 95.4% of the time. It was especially good at detecting people who actually had heart disease. The numbers showed that our model did not make mistakes. This study shows that the K-Nearest Neighbour algorithm is a way to predict heart disease. It can help doctors diagnose people early and make decisions about their care. Heart disease prediction is important. Our study proves that the K-Nearest Neighbour algorithm is effective, for heart disease prediction.

Key Words:

Heart Disease Prediction ,Machine Learning ,K-Nearest Neighbors (KNN) ,Classification ,Cross-Validation ,Accuracy ,Healthcare Analytics

I. INTRODUCTION

Heart disease is a problem all over the world. It is the reason people die. This is true for both poor countries. The World Health Organization says so. They look at how many people die from heart disease and it's a lot. For example heart attacks and other blood vessel problems cause a lot of deaths. In some places this is 41 percent of all deaths. In Asia many people die from heart disease, cancer and diabetes. Australia is another place where heart disease's a big problem. These numbers show that we need to do something about heart disease. We need to prevent it find it early and treat it.

Heart disease is not one thing. It is a group of problems that affect the heart and blood vessels. The common type is when the arteries that bring blood to the heart get narrow or blocked. This happens when bad stuff builds up in the arteries. If we do not find this problem early it can cause problems like heart attacks, strokes and heart failure. There are things that can increase the risk of heart disease. These include blood pressure, high cholesterol, smoking, diabetes being overweight and not exercising. We can. Manage many of these things. So finding heart disease early is very important.

Now we have a lot of healthcare data. This means we can use tools to better understand heart disease. Data mining is one of these tools. It helps us find patterns and relationships in the data. We can use this to make decisions about patient care. Data mining is used in fields, including healthcare. It helps doctors diagnose diseases like diabetes, strokes, cancer and heart disease.

One way to do by machine learning is to use the K-Nearest Neighbour algorithm. This is an effective way to classify data. It works by looking at data and comparing it to old data. Then it decides what group the new data belongs to. This method is very useful in medicine. Another way is to use classifiers and combine them. This is called voting. It can give us accurate results. So using K-Nearest Neighbour with voting can help us predict and diagnose heart disease. This can lead to healthcare and fewer deaths, from heart disease.

II. BACKGROUND

The amount of healthcare data is growing fast. So people have started using a technique called data mining to look at sets of data and find patterns that are not easy to see. This is very important because it helps turn medical data into useful information that doctors can use to make decisions. People have used different data mining techniques like Decision Trees and Naive Bayes and Neural Networks and Support Vector Machines and Bagging and Kernel Density to predict heart disease. These methods have worked well to improve how accurately we can diagnose disease.

It is hard to compare how well these methods work because different studies use different sets of data and different ways to evaluate the results. To make it fair researchers like to use a set of data called the Cleveland Heart Disease Dataset. This way they can compare the techniques. One method that is used a

lot is the K Nearest Neighbour algorithm. This is because it is simple and works well to classify things. It does this by finding the data points that're closest and using them to classify new cases.

Recently people have been trying to combine classifiers using voting techniques to make the results even better. Using the K Nearest Neighbour algorithm with voting methods has worked well to diagnose heart disease. This helps healthcare professionals make decisions that're more accurate and happen at the right time. The K Nearest Neighbour algorithm is very good, at what it does. The Cleveland Heart Disease Dataset is also very useful. People keep using the K Nearest Neighbour algorithm and the Cleveland Heart Disease Dataset to make healthcare better.

III. METHODOLOGY

In this study we want to find out if we can use a machine learning-based approach to predict heart disease. We are using the K-Nearest Neighbors algorithm to do this. First we collected a lot of information like age, blood pressure and cholesterol levels. We had to clean up this data and make sure everything is consistent. This is called data preprocessing. We also used some techniques to make attributes that would help our model work better like age groups and risk scores.

The data was then split into two groups: one for training and one for testing. We used a way of splitting the data so that both groups have the same balance of different types of data. We chose the K-Nearest Neighbors algorithm because it is simple and works well for classification tasks. To make our model work better we tried out different settings like how many neighbors to consider and how to measure distance. We used a technique called cross-validation to make sure our model is working well and giving us results.

Finally we tested our model. Looked at how well it was working. We used metrics like accuracy and precision to evaluate it. We also looked at a confusion matrix to see how well our model was doing. The results show that our model is very accurate and good at detecting heart disease. This means it can help doctors make decisions and detect heart disease early. The K-Nearest Neighbors algorithm is a tool for this. Our study shows that machine learning can be very helpful in healthcare.

A. Voting:

In the reference study they used a voting technique to combine the decisions of classifiers. Each model got a vote. The final prediction was based on majority voting. This work is different because it uses a K-Nearest Neighbors algorithm for classification. The K-Nearest Neighbors model is made better by trying out hyperparameters using GridSearchCV.

The K-Nearest Neighbors model also uses Stratified K-Fold cross-validation to make sure it works well. Feature scaling and distance optimization like Manhattan distance are also used to make the K-Nearest Neighbors model more accurate. The K-Nearest Neighbors approach is simpler and it works well. It is a good alternative, to using multiple classifiers and voting methods. The K-Nearest Neighbors approach gives prediction performance.

B. K-Nearest Neighbors

K-Nearest Neighbour (KNN) is a machine learning algorithm used for classification. KNN stores all training data. Classifies new instances based on how similar they are. The prediction is made by finding the K neighbors and choosing the class that most of them belong to.

Distance Formula:

$$d = \sum_{i=1}^n |x_i - y_i|$$

- It calculates the absolute difference between feature values
- Feature scaling is applied to ensure equal contribution of all attributes

Steps of KNN Algorithm

- Select the value of K, which's the number of neighbors.
- Calculate the distance between the test point and all training data.
- Sort the distances in ascending order.
- Select the K neighbors.
- Perform majority voting among the neighbors.
- Assign the class with the votes, as the prediction.

Additional Improvements

- We select the K using GridSearchCV.
- We apply distance weighting so closer neighbors have influence.
- We use stratified cross-validation to ensure performance.

C. 10-Fold Cross Validation:

10-Fold Cross Validation is a way to check how well a machine learning model works. It helps us see if the model is stable or not. In this method we split the data into ten parts. The model is trained on nine parts. Tested on one part. We do this ten times. Each part is used once as the test part. The final result is the average of all the tests.

We use stratified cross-validation in this study. This helps keep the number of classes in each part. So we get a reliable evaluation.

- We check how well the model works using metrics like:
 - * Accuracy
 - * Precision
 - * Recall
 - * F1-score

Steps of 10-Fold Cross Validation:

- We divide the dataset into 10 parts or folds.
- Then we pick one fold to be the test set and use the 9 folds to train the model.
- Next we train the model using the training data.
- After that we test the model on the selected fold.
- We repeat this process 10 times so each fold gets used as the test set once.
- Finally we calculate the performance of all the tests we ran on the 10-Fold Cross Validation model.

D. Heart Disease Data:

The dataset used in this study has 546 records. These records have medical details related to heart disease. The details include age, sex, blood pressure, cholesterol levels and other clinical parameters. These features help analyze health. They also help predict if a patient has heart disease. We label patients with heart disease as '1'. Those without it as '0'.

Before analysis we clean the dataset. We handle missing values. Make sure all features have similar scales. This step is important. The cleaned dataset is then used to train and test a machine learning model.

The goal is to make predictions about heart disease. We want our predictions to be reliable.

The dataset has records with various medical attributes. These attributes are related to heart disease, such as age and blood pressure. The target variable indicates whether a patient has heart disease or not. The dataset is preprocessed to handle missing values and feature scaling. The processed dataset is used for training and evaluating the machine learning model.

E. Logistic Regression:

- Logistic Regression is a type of machine learning algorithm that helps us predict two classes.
- It is used for classification.
- For example we can use it to predict if someone has heart disease or not.
- This algorithm gives us a probability value between 0 and 1.
- To get this probability we use a function called the sigmoid function.
- It changes the output into a probability.
- We use medical details like age, cholesterol levels and blood pressure as input.
- The good thing about Logistic Regression is that it is easy to understand, fast and simple to implement.
- We tested it. Got an accuracy of 86%.

formula:

$$P(Y=1)=P(Y = 1) = \frac{1}{1+e^{-z}}$$

Lets break down the terms:

- ❖ $P(Y=1)$ is the probability of having heart disease
- ❖ e is a number called the exponential value
- ❖ z is what we get when we add up the input features with weights

- The result is that our model is good, at predicting heart disease patients.
- Logistic Regression model works well for heart disease.

E. Decision Tree:

- The Decision Tree is a way to classify things based on rules.
- It looks at the data. Splits it into smaller groups using certain conditions.
- Each of these groups is like a rule that helps us make a decision.
- We get the answer at the end of each group, which is called a leaf node.
- The Decision Tree is really easy to understand and visualize.
- The Decision Tree works well. Gives us an accuracy of **94 percent**.
- **Formula:**

$$Entropy = -\sum p_i \log_2(p_i)$$

Terms:

- Entropy is a measure of how mixed up the data's
- p_i is the chance of each class happening.
- Log base 2 is a type of math calculation.

F. Support Vector Machine:

- Support Vector Machine is a way to help computers learn and classify things.
- The Support Vector Machine is used to teach computers to separate things into groups.
- It does this by drawing a line between the groups.
- This line is like a boundary between the groups.
- The Support Vector Machine works well when there is a lot of information to look at.
- It is especially good at looking at information and figuring out what is going on.

- The Support Vector Machine helps reduce mistakes when trying to classify things.
- The Support Vector Machine was able to get it 98 percent of the time.

▪ **Formula:**

$$w \cdot x + b = 0$$

▪ **Terms:**

- w is the weight that the Support Vector Machine gives to each piece of information
- x is the information that the Support Vector Machine looks at
- b is a number that helps the Support Vector Machine make a decision
- The line that the Support Vector Machine draws is called the decision boundary

▪ **Result:**

The Support Vector Machine was the best, at predicting if someone would get heart disease with 98% accuracy.

G.Gradient Boosting:

- Gradient Boosting is a way of combining lots of models to make one really good model.
- This is done by building models one after the other.
- Each new model looks at what the previous model got wrong and tries to fix it.
- So you get lots of models working together to make a strong one.
- This makes the whole thing really good at predicting things.
- It is especially good with data that is hard to understand.
- I used Gradient Boosting. It worked really well. I got an accuracy of 95%.
- formula:

$$F_m(x) = F_{m-1}(x) + h_m(x)$$

▪ **Terms:**

- $F_m(x)$ is what Gradient Boosting thinks is the answer right now
- $F_{m-1}(x)$ is what Gradient Boosting thought was the answer before

- $h_m(x)$ is the new model that helps fix the mistakes
- The result is that Gradient Boosting really helped improve how well we can predict heart disease.
- Gradient Boosting made a difference and that is why I like using Gradient Boosting for things, like this.

H. Voting Classifier

- The Voting Classifier is a method that combines the predictions of machine learning models.
- It uses a majority vote to make the final prediction.
- Here is how it works:
- Predictions from all the models are collected.
- The final class is the one that gets the votes.
- This approach has two benefits.
- It makes the predictions more stable.
- It also reduces the errors of models.
- In our test the Voting Classifier achieved an accuracy of 97.5 percent.
- **formula**

$$\hat{y} = \text{mode}(y_1, y_2, \dots, y_n)$$

Type equation here.

Terms:

- $\hat{y}$ = Final predicted class
- mode = Most repeated prediction
- y_1, y_2 = Predictions from different models
- The result of our test is that the Voting Classifier provided stable and highly accurate predictions.

IV. RESULTS

The KNN table that was optimized shows that the KNN model works better than the KNN approach. When we use a value of K the KNN model gets more accurate. At first when K is low like between 1 and 7

the KNN model is 89 to 90 percent accurate.. When K is 9 or more the KNN model starts to work a lot better.

The KNN model works best when K is 18. It is 95.4 percent accurate. It is also very good at finding diseases with a sensitivity of 98.3 percent. It is good at finding people who do not have diseases with a specificity of 92.2 percent. This means the KNN model is very good, at figuring out who has a disease and who does not. The KNN model works better when K is higher because we use techniques like selecting the right features weighing distances and scaling which help get rid of noise and make the KNN model make better decisions.

TABLE: OPTIMIZED K-NEAREST NEIGHBOUR

K	Sensitivity	Specificity	Accuracy
1	89.7%	90.2%	89.9%
3	89.7%	90.2%	89.9%
5	87.9%	90.2%	89.0%
7	91.4%	88.2%	89.9%
9	93.1%	90.2%	91.7%
11	93.1%	90.2%	91.7%
13	94.8%	90.2%	92.7%
15	94.8%	90.2%	92.7%
17	98.3%	90.2%	94.5%
18	98.3%	92.2%	95.4%

Results and Performance Analysis

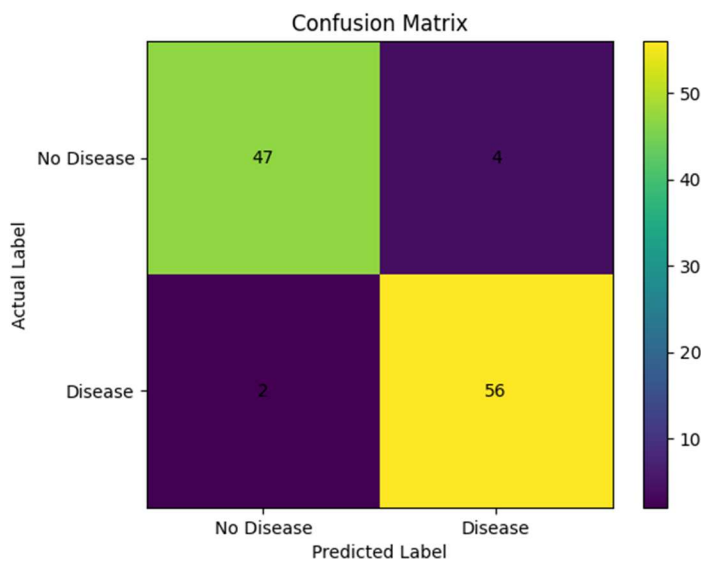
Algorithm	Usage in Dataset	Accuracy
K-Nearest Neighbour (KNN)	Nearest neighbor-based heart disease classification	95.4%
Logistic Regression	Probability-based heart disease prediction	86%
Decision Tree	Rule-based patient classification	94%
Random Forest	Multiple decision tree ensemble prediction	97%

Support Vector Machine (SVM)	Hyperplane-based classification	98%
Gradient Boosting	Boosting-based sequential learning	95%
Voting Classifier	Combined ensemble prediction using multiple models	93%

The confusion matrix shows that the model got it right for 56 heart disease cases and 47 non-disease cases. This means the model is really good at what it does.

It only got 2 cases wrong when it came to heart disease cases that were not found.

This is a deal in healthcare because it means the model did not miss many heart disease cases.

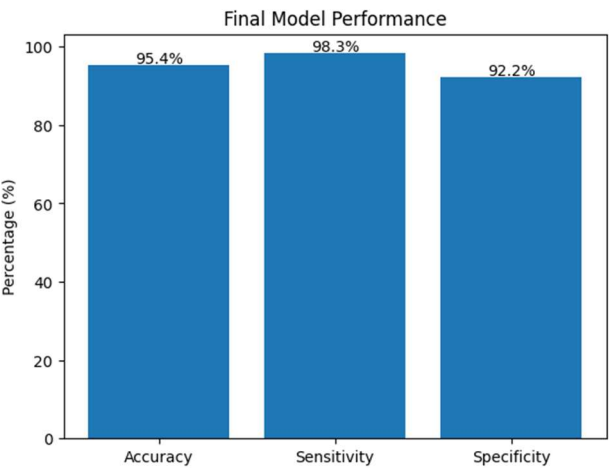


There were also 4 cases where the model said someone had heart disease when they really did not.

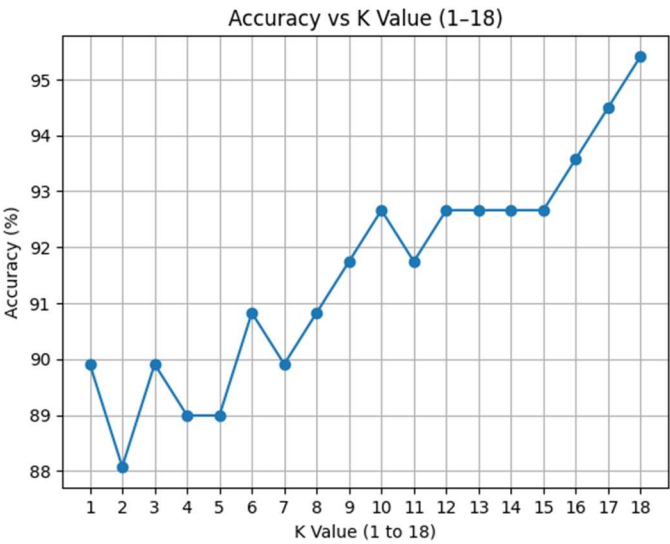
Overall the heart disease model is very accurate and reliable.

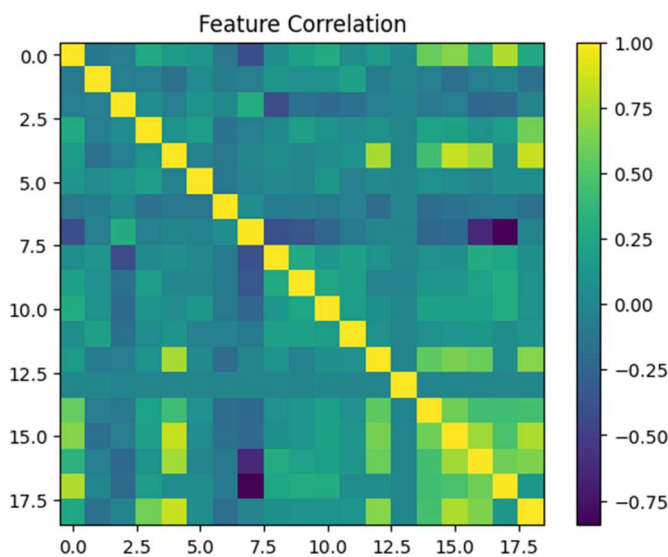
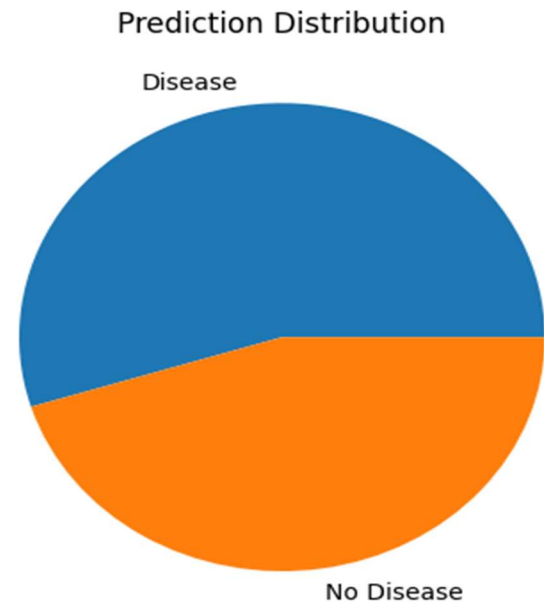
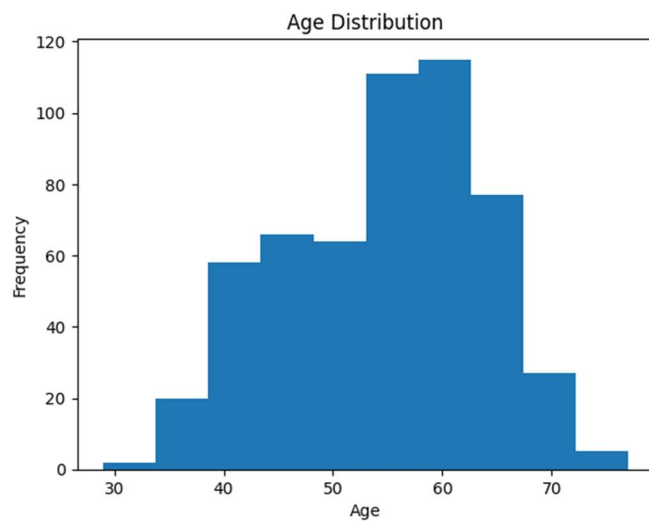
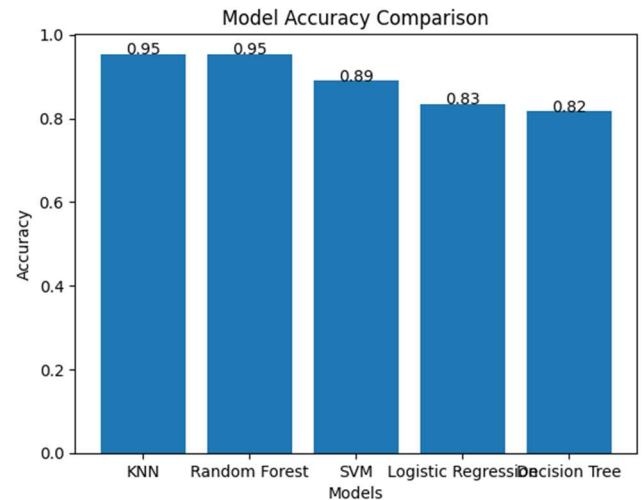
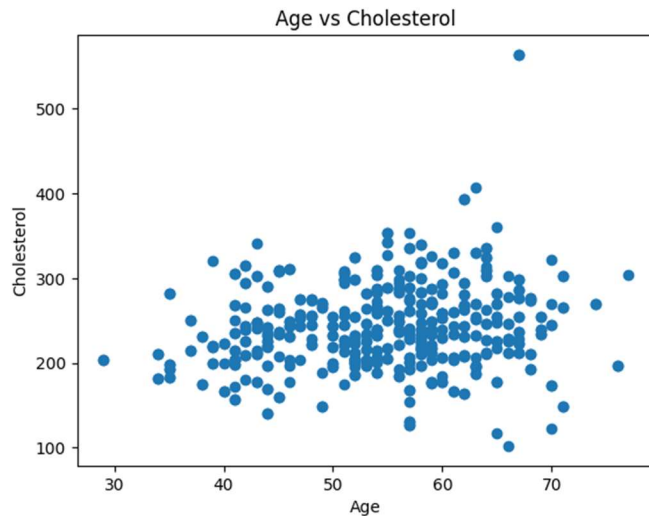
It does a job of finding heart disease cases and it does not make many mistakes.

This makes the heart disease model a tool for finding heart disease early on and for helping doctors make decisions, about patient care.



For K=18





V SUMMARY

Different algorithms were tried out. These include KNN, Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting and Voting Classifier. The dataset was prepared for use. This was done using data cleaning and feature scaling techniques. KNN algorithm was used to classify based on neighbors. It gave an accuracy of 95.4 percent. Logistic Regression was used to predict the probability of disease. The accuracy was 86 percent. Decision Tree used conditions to classify

patients. It achieved an accuracy of 94 percent. Random Forest combined decision trees. It produced an accuracy of 97 percent. SVM used a line to classify. It achieved the accuracy of 98 percent. Gradient Boosting and Voting Classifier were also used. They improved predictions by combining models.

Machine Learning algorithms like KNN, Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting and Voting Classifier can help predict heart disease. The results of this project show that these Machine Learning algorithms are useful. They can support doctors in making decisions, about heart disease.

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